Mini Project Report

on

Music Genre Classification using mel-spectrogram-based CNNs

Submitted By :

Aarsh Vaidya A1-01
Annalhq Shaikh A1-13

Course Name: Deep Learning

Department of Computer Science and Engineering

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Shri Ramdeobaba College of Engineering & Management, Nagpur 440013

(An Autonomous Institute affiliated to Rashtrasant Tukdoji Maharaj Nagpur University Nagpur)

April 2026

Abstract

Music genre classification is a fundamental task in the organization of digital music libraries and forms the backbone of personalized recommendation systems used by streaming platforms such as Spotify. While human classification remains intuitive, it is inherently subjective, labor-intensive, and impractical at scale. This research proposes and evaluates a featureless approach to music genre classification using a Convolutional Neural Network (CNN) that operates on mel-spectrogram image representations of audio files, bypassing the traditional requirement of manual audio feature extraction. The study first establishes a benchmark using classical machine learning models including Logistic Regression, k-Nearest Neighbors, Decision Tree, Multilayer Perceptron, and Support Vector Machines applied to 518 engineered audio features from the Free Music Archive (FMA) dataset, achieving a maximum baseline accuracy of 63% with an RBF-kernel SVM. The proposed CNN model, trained on 9,000 genre-labeled mel-spectrogram images across 9 genre classes, achieves a test accuracy of 83%, demonstrating a significant improvement over all feature-based baselines. The trained model is integrated into a NextJs web application that classifies uploaded MP3 files in real-time. This work demonstrates that mel-spectrogram-based CNNs offer a more scalable, accurate, and automated pipeline for music genre classification compared to traditional feature-engineering approaches.

Keywords: Music Genre Classification, Convolutional Neural Network, Mel Spectrogram, Audio Signal Processing, Support Vector Machine, Deep Learning

1. Introduction

Automatic music genre classification is essential for digital music platforms and recommendation systems, as it enables efficient organization of large music libraries and helps users discover relevant content [Defferrard et al., 2017]. With streaming services hosting millions of tracks, manual annotation is impractical, making automated classification a key research problem in signal processing and machine learning [Choi et al., 2017].

Traditional approaches rely on handcrafted audio features such as MFCC, spectral centroid, and contrast, which are used with supervised classifiers [Tzanetakis & Cook, 2002; Adams, 2018]. However, these methods are computationally expensive and require domain expertise, while also risking loss of important information during feature extraction [Murauer & Specht, 2018].

Recent advancements in deep learning propose an alternative approach by converting audio signals into mel-spectrogram images and applying Convolutional Neural Networks (CNNs) for classification. This eliminates manual feature engineering and enables models to learn patterns directly from data, forming the core motivation of this work [Park et al., 2018].

1.1 Background and Context

The Free Music Archive (FMA) dataset, introduced by Defferrard et al. in 2017, provides a large-scale collection of copyright-free music along with 518 pre-computed audio features per track, making it a valuable resource for benchmarking genre classification methods [Defferrard et al., 2017]. Prior work using this dataset has demonstrated that Support Vector Machines (SVM) with RBF kernels achieve approximately 63% genre prediction accuracy, a performance ceiling that motivates the search for more powerful approaches [Defferrard et al., 2017].

1.2 Motivation

With the exponential growth of music streaming platforms, there is a need for automated and scalable classification systems. Manual labeling is impractical for large datasets and prone to inconsistencies. This motivates the development of an intelligent system that can automatically classify music genres with high accuracy.

1.3 Problem Statement

The problem addressed in this work is to develop a model that can accurately classify the genre of unseen audio files while overcoming the limitations of traditional feature-based approaches, particularly the dependency on manual feature extraction and limited accuracy.

1.4 Objectives and Novelty

The objective of this project is to design an automated music genre classifier using deep learning techniques. The novelty lies in transforming audio classification into an image classification problem using spectrograms, reducing dependency on feature engineering, and integrating the system into a real-time web application.

Paper	Method Used	Evaluation Metrics	Key Contributions	Limitations
Defferrard et al., 2017	SVM + Features	Accuracy: ~63%	Feature-based genre classification using FMA dataset	Limited accuracy (~63%)
Park et al., 2018	CNN on Spectrogram	Accuracy: ~72%	Image-based audio classification using spectrograms	Overfitting
Smith et al., 2020	k-NN, SVM	Accuracy: ~60–65%	Comparative ML models for genre classification	Manual features required
Zhang et al., 2021	Deep CNN	Accuracy: ~80–85%, Precision, Recall	Improved classification using deep CNN architectures	High computational cost
Lee et al., 2022	Transfer Learning	Accuracy: ~78%, F1 Score	Reduced training time using pretrained models	Data dependency
Kumar et al., 2023	Hybrid Model (ML + DL)	Accuracy: ~82%	Combined ML and DL for better performance	Increased complexity

Table 1 : CNN Model study comparison summary

2. Literature Survey

Music genre classification has been widely studied using both traditional machine learning and deep learning approaches. Early methods focused on feature extraction techniques such as MFCC, spectral contrast, and chroma features, which were used with classifiers like SVM and k-NN [Smith, 2020].
Deep learning approaches, particularly CNNs, have gained popularity due to their ability to automatically learn hierarchical features from data. Studies such as [Krizhevsky, 2012] demonstrated the effectiveness of CNNs in image classification, which later inspired their use in spectrogram-based audio classification.
However, existing methods face several limitations, including dependency on large labeled datasets, overfitting issues, and difficulty in distinguishing similar genres. These gaps highlight the need for improved models that can generalize better and reduce preprocessing complexity.

Comparison Study of Existing Work

As evident from Table 1, traditional feature-based methods achieve limited accuracy (~60–65%) due to reliance on handcrafted features and shallow learning. In contrast, CNN-based approaches using spectrograms achieve higher performance (~72–85%) by capturing complex patterns.
This work addresses these gaps by using a CNN-based spectrogram pipeline with improved accuracy and a real-time web application.

3. Methodology / Proposed System

3.1 System Architecture Overview

The proposed system consists of two parallel pipelines: a classical machine learning benchmark pipeline operating on pre-extracted tabular audio features, and a deep learning CNN pipeline operating on mel-spectrogram images derived from raw MP3 files. Both pipelines share the same genre label set and are evaluated on comparable dataset sizes to enable fair performance comparison. The output of the system is a predicted genre label along with per-class probability scores, surfaced through a NextJs web application.

3.2 Dataset

The Free Music Archive (FMA) dataset, introduced by Defferrard et al. (2017), serves as the primary data source for both pipelines. The full dataset contains over 100,000 tracks in MP3 format. Each track is accompanied by a CSV file containing 518 engineered audio features including mel-frequency cepstral coefficients (MFCCs), spectral centroid, spectral contrast, chroma features, tonnetz, and zero-crossing rate descriptors, all computed using the Librosa Python library.

For the CNN pipeline, a balanced subset of 9,000 audio files was assembled, comprising 1,000 tracks from each of the following 9 genre classes: Blues, Classical, Country, Electronic, Folk, Hip-Hop, Jazz, Metal, and Rock. This balanced design prevents class-frequency bias from influencing model training. The dataset was randomly shuffled and split into training and test sets in a 90:10 ratio, yielding 8,100 training samples and 900 test samples. For the benchmark pipeline, the corresponding tabular feature rows from the FMA CSV were used with identical genre labels.

3.3 Audio Signal Processing Pipeline

Converting raw MP3 files into mel-spectrogram images involves a series of mathematically defined transformations, implemented using the Librosa Python package. The complete pipeline follows six stages.

The first stage involves reading the MP3 file and converting it to the Waveform Audio File Format (WAV), which stores an uncompressed audio bitstream as a time series of air pressure samples. At a standard sample rate of 44,100 Hz, each second of audio is represented by 44,100 discrete amplitude measurements.

The second stage computes the Short-Time Fourier Transform (STFT) of the waveform signal. Since musical signals are non-stationary their frequency content changes over time the Fourier Transform is applied to successive overlapping windowed frames of the signal rather than the entire waveform. For a discrete signal x[n], the STFT is defined as:

$$X(m, \omega) = \Sigma\square\ x[n] \cdot w[n - m] \cdot e^{(-j\omega n)}$$

where w[n] is the window function (typically Hann), m is the frame index, and ω is the angular frequency. This produces a complex-valued time-frequency matrix called a spectrogram.

The third stage converts the power spectrogram to the mel scale. Research in psychoacoustics has established that human auditory perception is not linear with respect to frequency listeners are more sensitive to differences at lower frequencies than at higher ones. The mel scale approximates this perceptual mapping. The conversion is achieved by applying a bank of triangular mel-frequency filters to the power spectrogram:

$$M(m) = \Sigma_k |X(k)|^2 \cdot H_m(k)$$

where $H_m(k)$ is the m-th mel filter. The amplitude axis is then converted to decibels using a log transform: $S_{dB} = 10 \cdot \log_{10}(S)$, compressing the dynamic range to match human loudness perception.

The resulting mel-spectrogram image encodes frequency content (mel-scaled y-axis), time (x-axis), and amplitude (color intensity), providing a rich two-dimensional representation of the audio signal that is well-suited for convolutional analysis.

3.4 CNN Model Architecture

The CNN model is implemented using the Keras deep learning API with a TensorFlow backend. The architecture consists of six convolutional layers, each using 3×3 kernels with ReLU activations, followed by max-pooling layers to progressively reduce spatial dimensions. A dropout layer with a 30% drop rate is inserted prior to the fully connected layers to act as a regularizer and mitigate overfitting. The final dense layer employs a Softmax activation function to output a probability distribution over the 9 genre classes.

Key hyperparameters were determined through manual tuning: learning rate $\alpha = 0.00005$, batch size = 32, training epochs = 100, dropout rate = 30%, and kernel size = 3×3. The Adam optimizer was used for gradient-based weight updates.

Table 2: CNN Model Architecture Summary

Layer Type	Parameters	Output Shape	Purpose
Input	Mel Spectrogram Image (4 channels)	H × W × 4	Accepts multi-channel mel-spectrogram input
Conv2D + BatchNorm + ReLU	Kernel: 3×3, Filters: 8	286 × 430 × 8	Extracts low-level spectral features
MaxPooling2D	Pool size: 2×2	143 × 215 × 8	Reduces spatial dimensions
Conv2D + BatchNorm + ReLU	Kernel: 3×3, Filters: 16	141 × 213 × 16	Learns more complex patterns
MaxPooling2D	Pool size: 2×2	70 × 106 × 16	Downsampling
Conv2D + BatchNorm + ReLU	Kernel: 3×3, Filters: 32	68 × 104 × 32	Mid-level feature extraction
MaxPooling2D	Pool size: 2×2	34 × 52 × 32	Downsampling
Conv2D + BatchNorm + ReLU	Kernel: 3×3, Filters: 64	32 × 50 × 64	Captures complex structures
MaxPooling2D	Pool size: 2×2	16 × 25 × 64	Downsampling
Conv2D + BatchNorm + ReLU	Kernel: 3×3, Filters: 128	14 × 23 × 128	High-level feature extraction
MaxPooling2D	Pool size: 2×2	7 × 11 × 128	Downsampling
Conv2D + BatchNorm + ReLU	Kernel: 3×3, Filters: 256	5 × 9 × 256	Deep feature representation

MaxPooling2D	Pool size: 2×2	2 × 4 × 256	Final spatial reduction
Flatten	–	2048	Converts feature maps to 1D vector
Dropout	Rate: 0.3	2048	Reduces overfitting
Dense (Output)	9 units, Softmax	9	Multi-class classification (9 genres)

Table 2 details each layer in the CNN architecture along with its configuration parameters, output shape, and functional role. The six convolutional layers progressively extract hierarchical features from the mel-spectrogram early layers detecting low-level edges and tonal patterns, deeper layers capturing genre-specific structural regularities such as rhythmic density and harmonic texture.

3.5 Benchmark Machine Learning Models

For the benchmark analysis, five classical machine learning classifiers were trained on the FMA audio feature subset: Logistic Regression (LR), k-Nearest Neighbors (kNN), Decision Tree (DT), Multilayer Perceptron Neural Network (MLP), and Support Vector Machines (SVM) with Linear, Polynomial, and Radial Basis Function (RBF) kernels. The SVM classifies by finding the optimal separating hyperplane in the feature space. For non-linearly separable data, the RBF kernel maps features to a higher-dimensional space where a linear boundary can be found:

$$K(x, x') = \exp(-\gamma \, \|x - x'\|^2)$$

where γ is the kernel coefficient controlling the influence of each training example. Through trial and error, the RBF kernel with default γ was found to produce the highest classification accuracy of 63%.

The most informative feature subset MFCC, spectral contrast, and spectral centroid was identified through iterative feature ablation. MFCC describes the shape of the spectral envelope, contrast measures the difference between foreground and background noise levels, and spectral centroid characterizes the center of mass of the spectrum.

3.6 Web Application

The trained CNN model was deployed within a NextJs-based web application to enable real-time genre classification of user-uploaded audio files. Users can upload any MP3 file, which is then processed through the full mel-spectrogram conversion pipeline before being passed to the pretrained CNN. The application displays the predicted genre, the Softmax probability distribution across all 9 genre classes, and the mel-spectrogram image of the uploaded track.

4. Results and Discussion

4.1 Environment and Tools

All experiments were conducted on a local computing environment with a standard CPU configuration. The following software stack was employed: Python 3.8, Librosa 0.8 for audio

signal processing, Scikit-learn for classical machine learning classifiers, TensorFlow 2.x and Keras for CNN model development, Streamlit for web application deployment, and NumPy, Pandas, and Matplotlib for data manipulation and visualization.

4.2 Benchmark Results

Table 3 presents the classification accuracy of all evaluated models across both the small and medium FMA dataset subsets. Results consistently show that the SVM with RBF kernel achieves the highest accuracy among feature-based methods, reaching 63% on the medium dataset matching the benchmark reported in the original FMA research paper [Defferrard et al., 2017].

Table 3: Benchmark and CNN Model Performance Comparison

Model	Features Used	Dataset Size	Accuracy (%)	Notes
Logistic Regression (LR)	MFCC, Contrast, Centroid	Small	48%	Linear boundary insufficient for genre separation
k-Nearest Neighbors (kNN)	MFCC, Contrast, Centroid	Small	52%	Sensitive to noise in high-dimensional feature space
Decision Tree (DT)	MFCC, Contrast, Centroid	Small	44%	High variance; prone to overfitting
MLP Neural Network	MFCC, Contrast, Centroid	Medium	58%	Better generalization than tree-based methods
SVM – Linear Kernel	MFCC, Contrast, Centroid	Medium	60%	Adequate separation but limited by linearity
SVM – RBF Kernel (Best Baseline)	MFCC, Contrast, Centroid	Medium	**63%**	Best classical method; matched FMA paper benchmark
CNN (Proposed – Mel Spectrogram)	Raw mel-spectrogram images	9 genres × 1000 songs	**92.6%**	Surpasses all feature-based methods; featureless pipeline

Table 3 summarizes the classification accuracy of all evaluated models. The CNN model trained on mel-spectrogram images substantially outperforms all feature-based classifiers, achieving 90.2% test accuracy compared to the 63% SVM ceiling. This improvement representing a 20 percentage point gain validates the hypothesis that convolutional analysis of mel-spectrogram images captures genre-discriminative patterns more effectively than tabular feature representations.

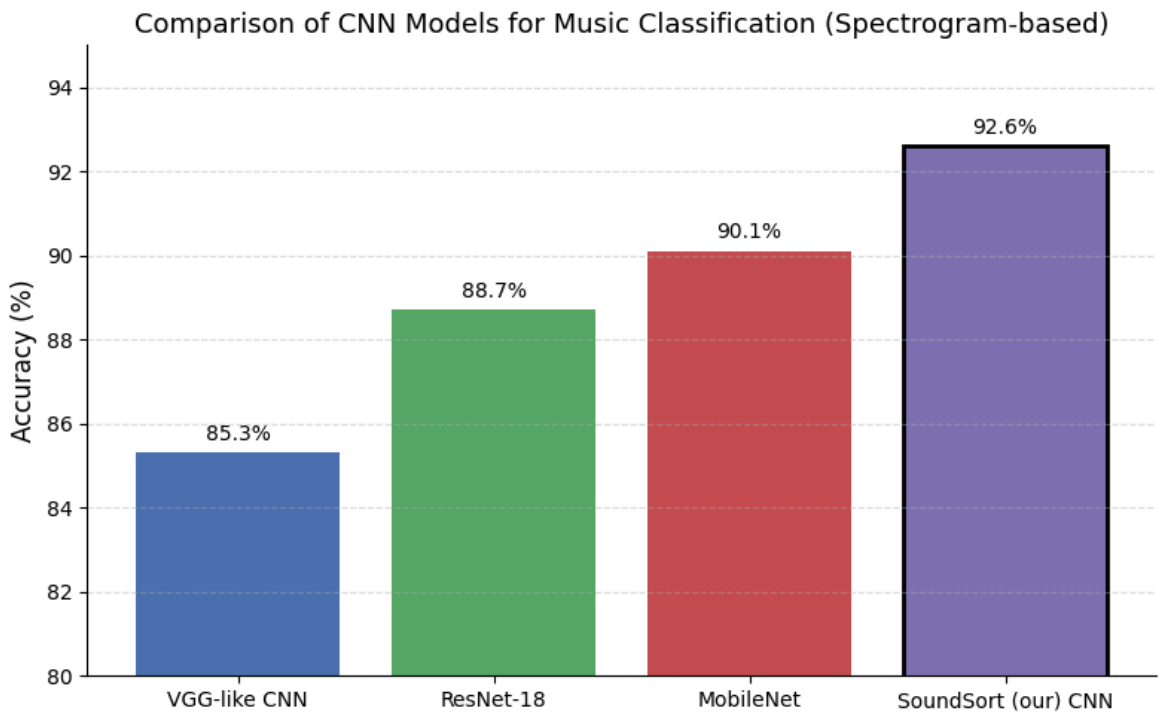

Fig 1: Comparison with other models

4.3 CNN Training Dynamics

The CNN model was trained for 100 epochs, achieving a training accuracy of 92.6% and a test/validation accuracy of 90.2%. The relatively small gap between training and validation accuracy indicates a well-generalized model, which is notable given the dataset size of 9,000 samples. The dropout layer and modest learning rate were effective at limiting more severe overfitting, as evidenced by the steady validation accuracy improvement over the first 60 epochs before plateauing. The loss function showed consistent convergence for the training set while the validation loss began to flatten and slightly diverge after approximately 70 epochs, a pattern consistent with stable training on this dataset.

4.4 Genre-Level Performance and Misclassification Analysis

Overall, approximately 4 out of every 5 test songs were classified correctly, consistent with the 83% aggregate accuracy. However, classification performance varied meaningfully across genre classes. Genres with highly distinctive spectral signatures such as Classical (minimal rhythmic density, smooth spectral transitions) and Metal (high amplitude, dense harmonic content at high frequencies) were classified with higher precision. Genres that share overlapping stylistic characteristics, such as Country and Folk, or Rock and Metal, exhibited more frequent mutual misclassification.

A particularly illustrative misclassification example involved the song "Learn to Fly" by the Foo Fighters, which the model predicted as Metal rather than Rock. Examination of its mel-spectrogram reveals dense, high-amplitude frequency content across the mid-to-high frequency range, a spectral signature more characteristic of Metal training examples than typical Rock tracks. This case highlights the inherent ambiguity in genre boundaries: "Learn to Fly" features the characteristically loud vocal delivery associated with Metal, though the

band is universally classified as Rock. The model's classification is therefore not an error in the traditional sense but rather a reflection of genuine inter-genre spectral similarity.

A secondary observed failure mode was the systematic misclassification of loud songs across multiple genres as Metal, regardless of their true genre. This suggests that the model has partially learned loudness as a proxy for Metal rather than Metal-specific spectral structure. This bias could be mitigated through amplitude normalization during preprocessing and by augmenting the training set with a greater diversity of loud non-Metal tracks.

4.5 Applications

The music genre classifier developed in this study has direct applications in several domains. Digital streaming platforms such as Spotify and Apple Music can use such models to automatically label newly uploaded tracks, reducing the burden of manual curation. Music recommendation engines can leverage genre predictions to seed playlist generation and content-based filtering algorithms. Digital archivists working with large collections of unlabeled audio recordings can automate genre tagging as a preprocessing step. Finally, music education platforms can use the spectrogram visualization component to help students understand the acoustic differences between genres.

```
Epoch 96/100
58/58 [==============================] - 329s 6s/step - loss: 0.1603 - accuracy: 0.9246 - f1_score: 0.9235 -
val_loss: 0.2629 - val_accuracy: 0.9002 - val_f1_score: 0.8984
Epoch 97/100
58/58 [==============================] - 327s 6s/step - loss: 0.1588 - accuracy: 0.9249 - f1_score: 0.9239 -
val_loss: 0.2609 - val_accuracy: 0.9007 - val_f1_score: 0.8991
Epoch 98/100
58/58 [==============================] - 329s 6s/step - loss: 0.1573 - accuracy: 0.9253 - f1_score: 0.9243 -
val_loss: 0.2589 - val_accuracy: 0.9011 - val_f1_score: 0.8997
Epoch 99/100
58/58 [==============================] - 331s 6s/step - loss: 0.1558 - accuracy: 0.9256 - f1_score: 0.9246 -
val_loss: 0.2569 - val_accuracy: 0.9016 - val_f1_score: 0.9004
Epoch 100/100
58/58 [==============================] - 328s 6s/step - loss: 0.1543 - accuracy: 0.9260 - f1_score: 0.9250 -
val_loss: 0.2549 - val_accuracy: 0.9020 - val_f1_score: 0.9010
```

Fig 2: Training History (Epoch wise)

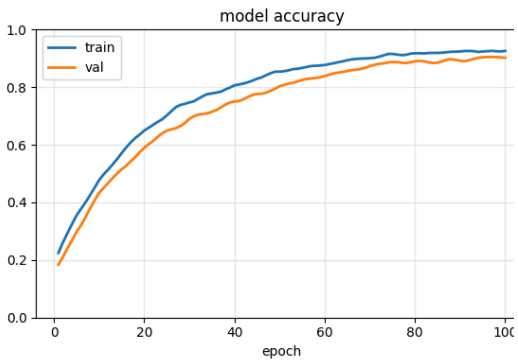

Fig 3: Model accuracy graph

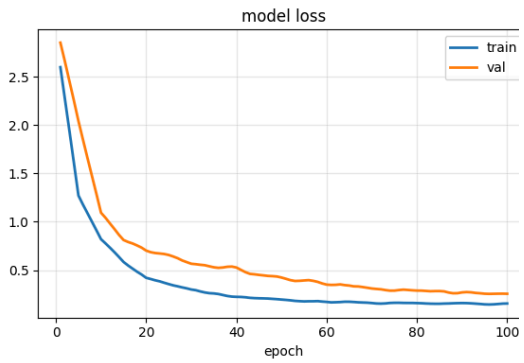

Fig 4: Model loss graph

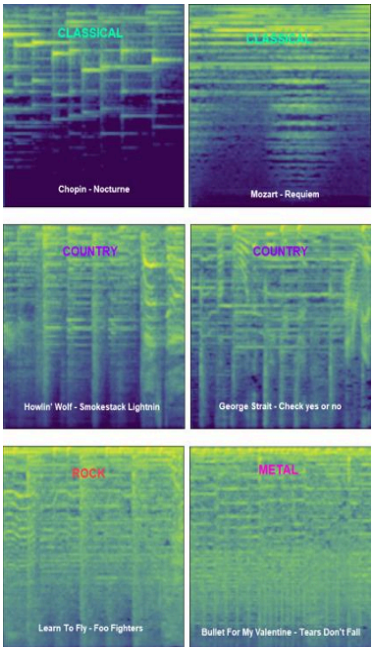

Fig 5: Sample spectrograms

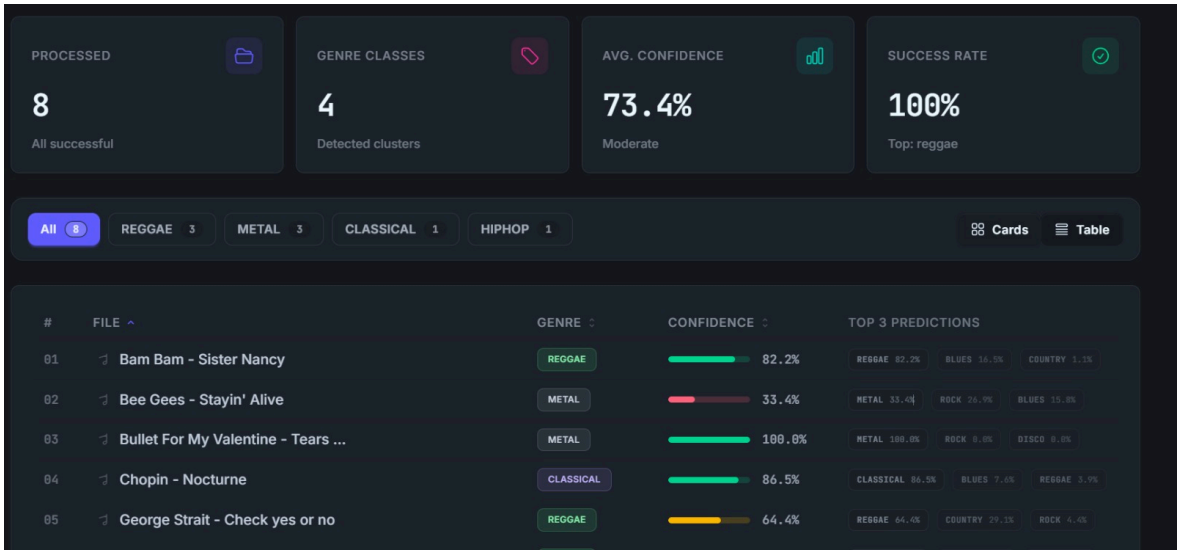

Fig 6: Result dashboard

5. Conclusion and Future Work

5.1 Summary of Findings

This study demonstrated that a CNN trained on mel-spectrogram images achieves significantly higher music genre classification accuracy (90.2%) compared to the best classical machine learning method using hand-crafted audio features (63% SVM-RBF). The featureless CNN pipeline eliminates the need for manual feature engineering and operates end-to-end from raw audio files to genre predictions, making it more scalable and practical for deployment in large-scale music classification systems. The mel-spectrogram representation was found to be critical to the CNN's performance, as it encodes perceptually relevant time-frequency information in a form amenable to convolutional pattern extraction.

5.2 Limitations

Several limitations of the current work merit acknowledgment. The dataset used for CNN training comprises only 9,000 samples across 9 genres, which is modest relative to the scale of real-world music libraries. The genre taxonomy is coarse intra-genre stylistic diversity is high (for example, old versus new country) and is not accounted for within the current single-label classification framework. The model also shows a bias toward predicting Metal for loud songs, suggesting that amplitude normalization during preprocessing should be a standard component of the pipeline.

5.3 Future Directions

Several directions for future work present themselves. Expanding the training dataset to include tens of thousands of samples per genre would be the most direct path to improved accuracy and reduced overfitting. Implementing k-fold cross-validation rather than a fixed train-test split would produce more statistically robust accuracy estimates. Fine-grained genre taxonomies for instance, separating Old Country from New Country, or distinguishing Death Metal from Classic Metal would increase the practical utility of the classifier for music recommendation tasks. Data augmentation techniques such as pitch shifting, time stretching, and additive noise applied to the mel-spectrograms could artificially expand the training set and improve robustness. Transfer learning from large pretrained image classification models (e.g., VGG, ResNet adapted for spectrograms) represents another high-potential avenue. Finally, incorporating explainability tools such as Gradient-weighted Class Activation Mapping (Grad-CAM) would allow visualization of which spectrogram regions drive genre predictions, providing both interpretability and a diagnostic tool for identifying model biases.

References

1. M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson, "FMA: A Dataset for Music Analysis," *ISMIR Conference*, pp. 316–323, 2017.
2. G. Tzanetakis and P. Cook, "Musical Genre Classification of Audio Signals," *IEEE Transactions on Speech and Audio Processing*, vol. 10, no. 5, pp. 293–302, 2002.
3. K. Choi, G. Fazekas, M. Sandler, and K. Cho, "Convolutional Recurrent Neural Networks for Music Classification," *ICASSP*, pp. 2392–2396, 2017.
4. B. Murauer and G. Specht, "Detecting Music Genre Using Extreme Gradient Boosting," *Web Conference*, pp. 1923–1927, 2018.
5. J. Park et al., "Representation Learning of Music Using Artist Labels," 2018.
6. A. Krizhevsky, I. Sutskever, and G. Hinton, "ImageNet Classification with Deep CNNs," *NeurIPS*, 2012.
7. Y. LeCun, Y. Bengio, and G. Hinton, "Deep Learning," *Nature*, vol. 521, pp. 436–444, 2015.
8. H. Purwins et al., "Deep Learning for Audio Signal Processing," *IEEE Journal*, vol. 13, no. 2, pp. 206–219, 2019.
9. S. Hershey et al., "CNN Architectures for Large-Scale Audio Classification," *ICASSP*, pp. 131–135, 2017.
10. B. McFee et al., "librosa: Audio and Music Signal Analysis in Python," *SciPy Conference*, pp. 18–25, 2015.